Introduction to Artificial Intelligence (AI)

Linear Regression

Salymbekov University
Miss Aliia Beishenalieva
aliya.beiwenalieva@gmail.com

Linear Regression

Predicting House Prices

1. Data Collection
A dataset of house prices is collected with features (input variables) like:

- Number of bedrooms
- Size of the house (square footage)
- Location
- Age of the house
- Number of bathrooms, etc.

Each house in the dataset has a **label** (target output), which is the actual price of the house.

2. Training the Model
The dataset is split into a **training set** and a **test set**.
A supervised learning algorithm, such as **linear regression**, is used to learn a mapping from the input features to the house price by minimizing the error between predicted and actual prices in the training data.

3. Model Prediction
After training, the model can be used to predict house prices for new houses based on their features.

4. Evaluation
The model's performance is evaluated using metrics like Mean Squared Error (MSE) on the test set to measure how well the model generalizes to new, unseen data.

Linear Regression

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

# Example dataset with house features (number of rooms, house size in sqft, and age of ho
data = {
    'Rooms': [3, 4, 2, 5, 4, 3, 5, 2, 4, 3],
    'Size (sqft)': [1500, 2000, 850, 2500, 1800, 1300, 2400, 900, 1750, 1400],
    'Age (years)': [10, 5, 20, 2, 7, 15, 3, 25, 8, 12],
    'Price ($)': [300000, 400000, 200000, 500000, 370000, 280000, 480000, 220000, 350000,
}

# Convert data to pandas DataFrame
df = pd.DataFrame(data)

# Features (independent variables): Rooms, Size (sqft), Age (years)
X = df[['Rooms', 'Size (sqft)', 'Age (years)']]
```

Linear Regression

```
# Target (dependent variable): Price
y = df['Price ($)']

# Split the data into training and test sets (80% training, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Create a linear regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Make predictions on the test set
y_pred = model.predict(X_test)

# Calculate the Mean Squared Error (MSE) for model evaluation
mse = mean_squared_error(y_test, y_pred)
```

Linear Regression

```python
# Print the results
print(f"Predicted Prices: {y_pred}")
print(f"Actual Prices: {y_test.values}")
print(f"Mean Squared Error: {mse}")

# Print model coefficients (weights) and intercept
print(f"Model Coefficients: {model.coef_}")
print(f"Model Intercept: {model.intercept_}")
```

Linear Regression

Dataset: The dataset has 3 features (Rooms, Size (sqft), Age (years)) and a target variable (Price ($)).
Data Preparation: We split the data into features (X) and the target variable (y), and then into training and test sets (80% for training, 20% for testing).
Model: We use the Linear Regression model from scikit-learn.
Training: The model is trained using the fit() method with the training data.
Prediction: We predict the house prices for the test set using predict().
Evaluation: The model's performance is evaluated using Mean Squared Error (MSE), which gives us an idea of how close the predictions are to the actual house prices.
Coefficients and Intercept: We print the learned weights (coefficients) for each feature and the intercept.

Output:
You will see the predicted prices, actual prices, the mean squared error, and the learned coefficients of the model.

Classification

Yes or No